

An aerial photograph of the University of British Columbia Okanagan campus. The image shows several large, modern brick buildings with flat roofs, some featuring solar panels. The campus is surrounded by lush green trees and lawns. In the background, there are rolling hills and mountains under a clear blue sky with a few wispy clouds. The lighting suggests it's either early morning or late afternoon, with long shadows cast across the landscape.

Special Topics: Applied Computer Vision and Deep Learning

UBCO Master of Data Science – DATA 589





A Limitation of CNNs

CNNs are built on **locality**:

- Nearby pixels are strongly correlated

The limitation

- Each convolution sees **only a local neighborhood**
- Early layers have **small, disconnected receptive fields**
- No direct mechanism to share **global context**
- CNNs are excellent at learning *local patterns*, but global understanding is not native to their design.

The only way CNNs can share global information is:

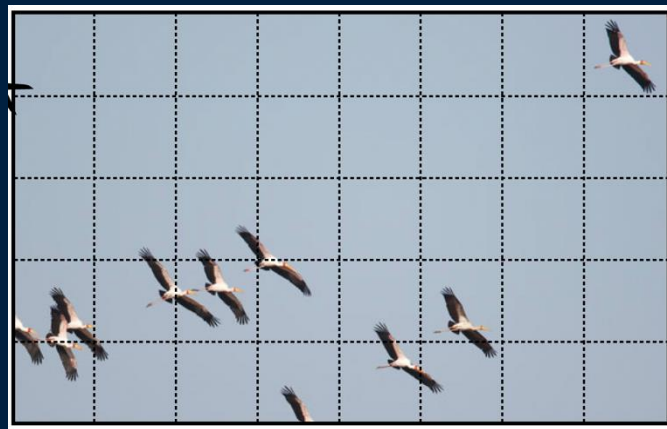
By stacking many layers — gradually increasing the receptive field

Or by using very large kernels — which is inefficient and breaks locality

- **CNNs aggregate information progressively**, but they are not designed to **model long-range dependencies explicitly**.

How many birds are in this image?

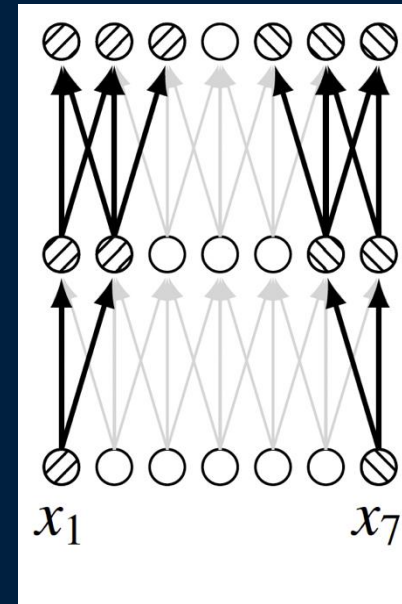
Is the top right bird the same species as the bottom left bird?



Isola + Foundations of Computer Vision

A Limitation of CNNs

- Far apart image patches do not interact.
- There are no neurons that are connected to x_1 and x_7 . They are too far away from each other.



The Idea of Attention

- **Attention** is a mechanism for processing **global information efficiently**

Instead of treating all parts of an image equally, the model **focuses on the most relevant regions** for the task

- **Motivation from human vision:**

Humans do not look at a whole scene at once

Our eyes move and attend to **salient objects**

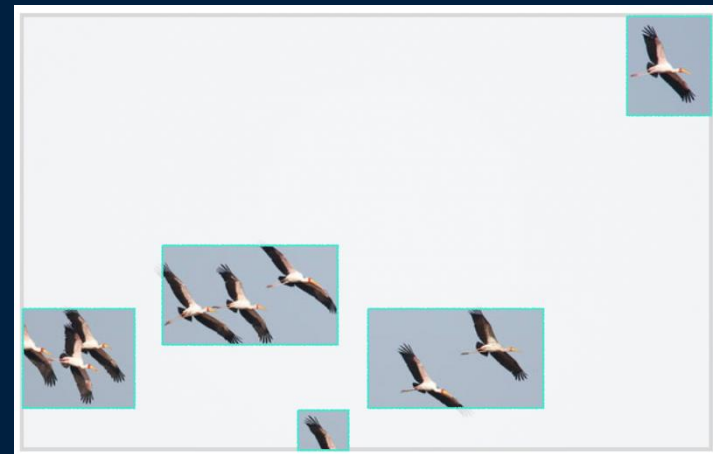
If asked about the color of a car, we focus on the **car**, not the background

- **Attention in neural networks:**

Neurons in **layer L+1 selectively attend to neurons in layer L**

The network decides **which features are important** for its output.

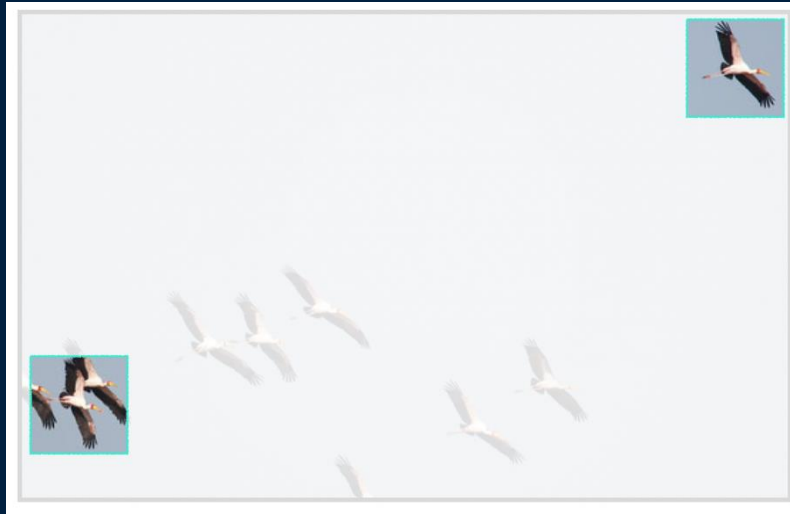
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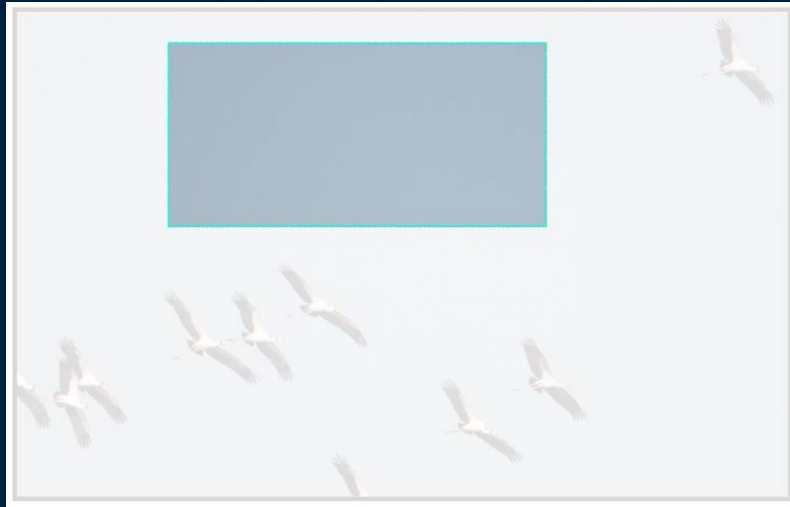
The Idea of Attention

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The Idea of Attention

Is the top right bird the same species as the bottom left bird?



Three Key Architecture of Innovations

- Tokens
- Attention
- Positional Codes

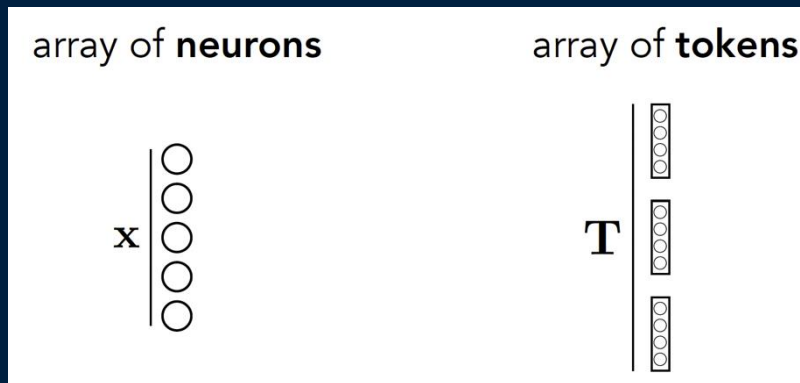
Question

Question: What is one major limitation of CNNs that motivates the use of attention?

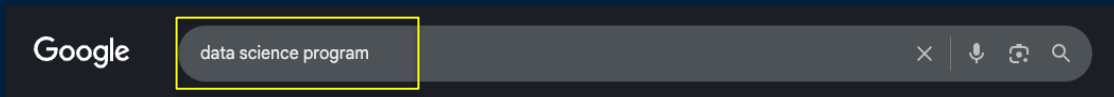
- A. CNNs cannot learn visual patterns
- B. CNNs cannot process RGB images
- C. CNNs mainly model local neighborhoods and do not explicitly capture long-range dependencies
- D. CNNs always require more data than transformers
- E. CNNs cannot be trained with backpropagation

A New Data Type: Tokens

- **A Token is just a vector of neurons.**
- But the connotation is that a token is an encapsulated bundle of information; with transformers we will operate over tokens rather than over neurons.



Understanding Attention with Search



Query(Q)

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Key(K1)

 GeeksforGeeks
<https://www.geeksforgeeks.org> > c > c-programming-ex...

C Programming Examples

Dec 27, 2025 — We have compiled over 100 C programming examples across various categories, including **basic C programs**, Fibonacci series, strings, arrays, base conversions, ... [Read more](#)

Key(K2)

 Coursera
<https://www.coursera.org> > browse > data-science

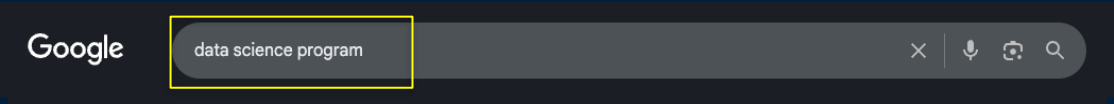
Data Science Online Courses

You'll gain skills in statistical analysis, machine learning, Python or R **programming**, **data** visualization, and database management. [Read more](#)

Key3(K3)



Understanding Attention with Search



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Query(Q)

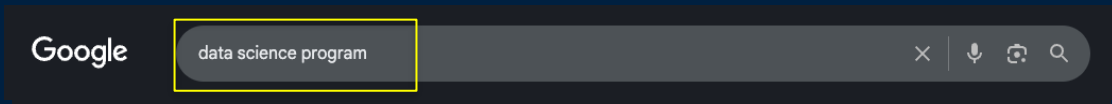
Key(K1)

Key(K2)

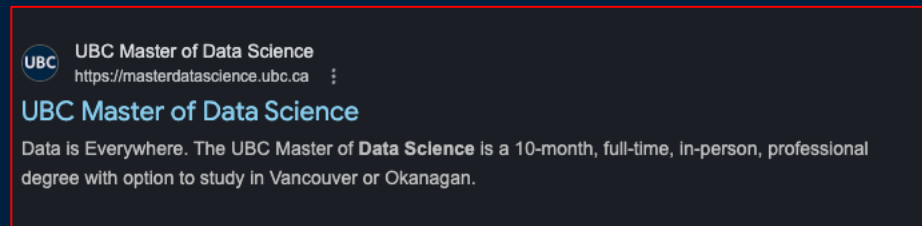
Key3(K3)

I. **Compute attention mask**: how similar is each key to the desired query

Understanding Attention with Search



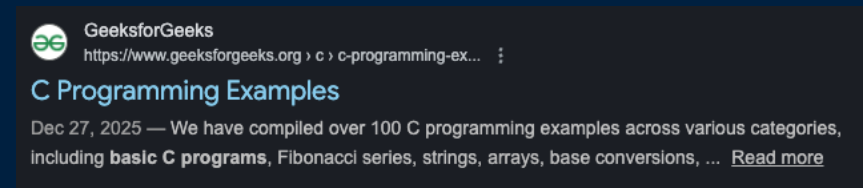
Query(Q)



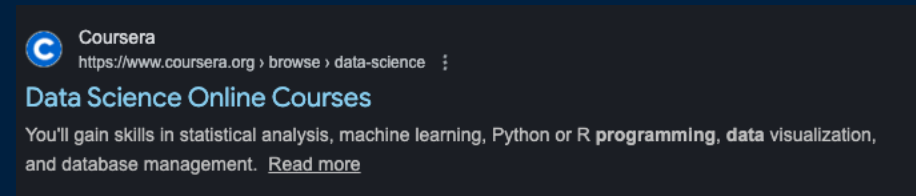
Key(K1)

Value (V)

II. Extract values based on attention:
 return the values with highest attention



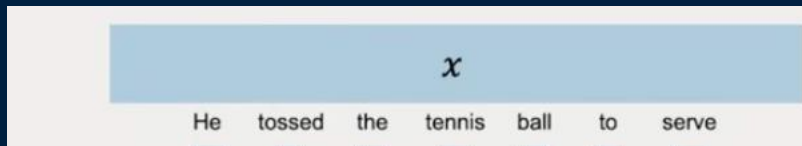
Key(K2)



Key3(K3)

Learning Self-Attention with NN

Goal: identify and attend to most important features in input.

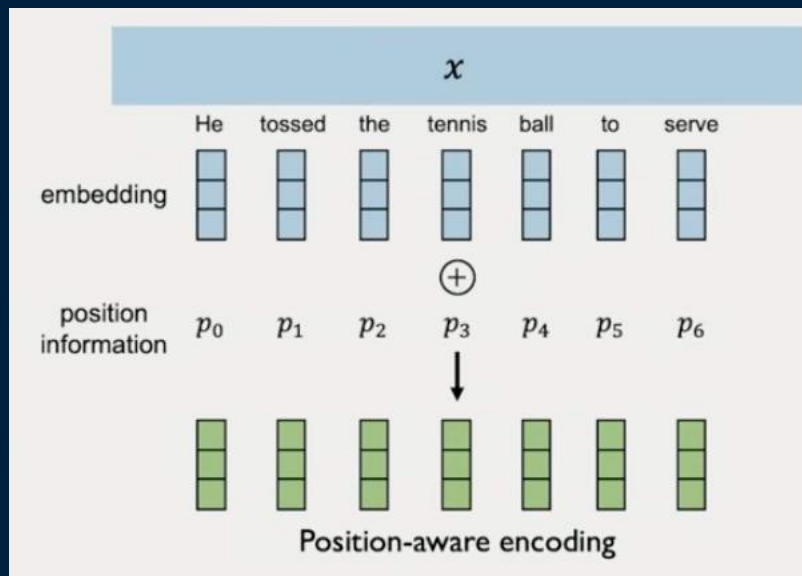


Learning Self-Attention with NN

Goal: identify and attend to most important features in input.

1. Encode position information

Data is fed in all at once! Need to encode position information to understand order.



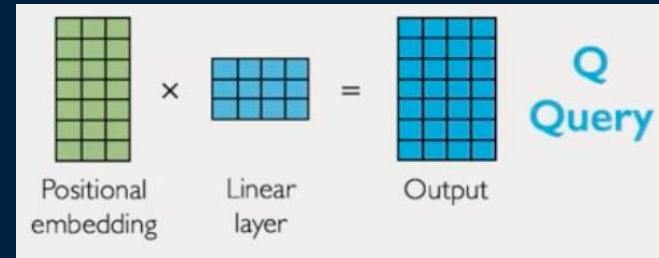
Learning Self-Attention with NN

1. Encode position information
2. Extract query, key, value for each token



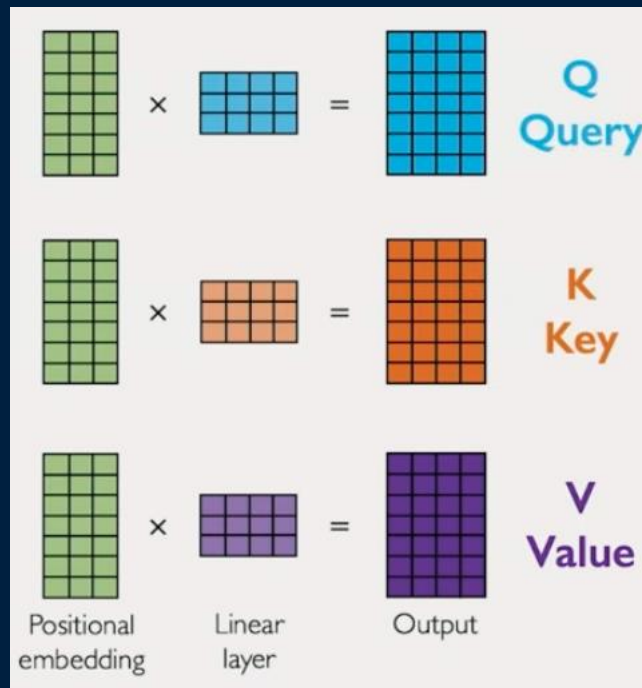
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Learning Self-Attention with NN

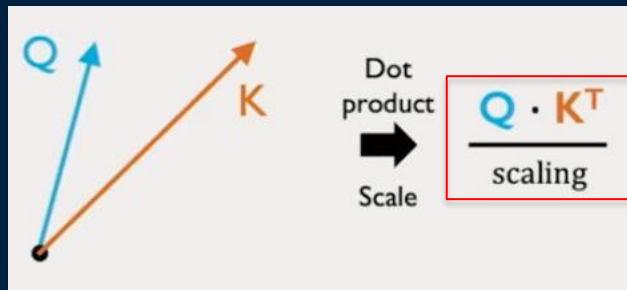
1. Encode position information
2. Extract query, key, value for each token
3. Compute attention weights?

Learning Self-Attention with NN

1. Encode position information
2. Extract query, key, value for each token
3. Compute attention weights?

Attention score: compute pairwise similarity between each query and key

How to compute the similarity between two sets of features?



Similarity metrics

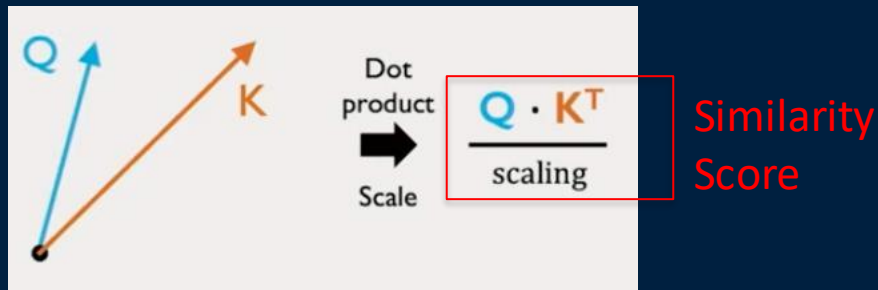
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Attention score: compute pairwise similarity between each query and key

Similarity Score: How relevant is this token (Key) to what the query looking for (Query)

How to compute the similarity between two sets of features?



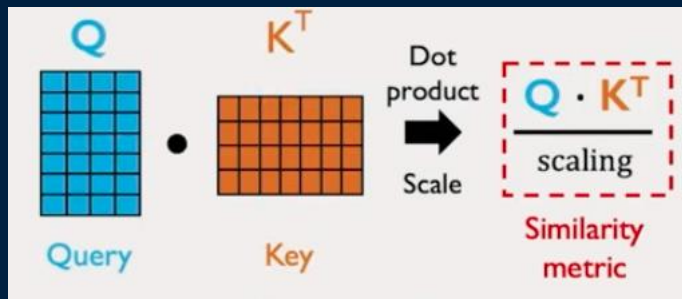
★ Learning Self-Attention with NN

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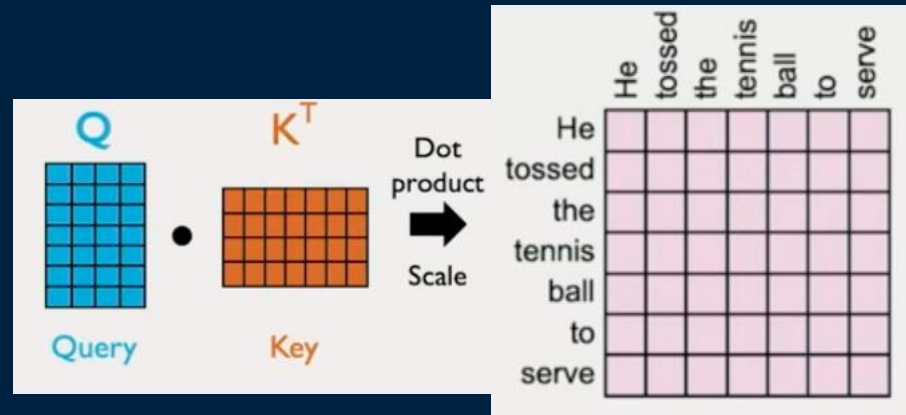


Similarity Score: How relevant is this token (Key) to what the query looking for (Query)

Learning Self-Attention with NN

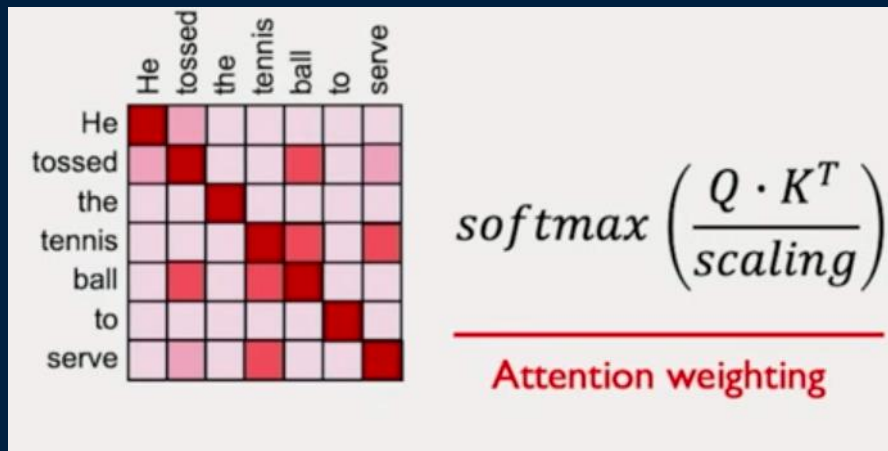
1. Encode position information
2. Extract query, key, value for each token
3. Compute attention weights

Similarity Score: How similar is Key Query



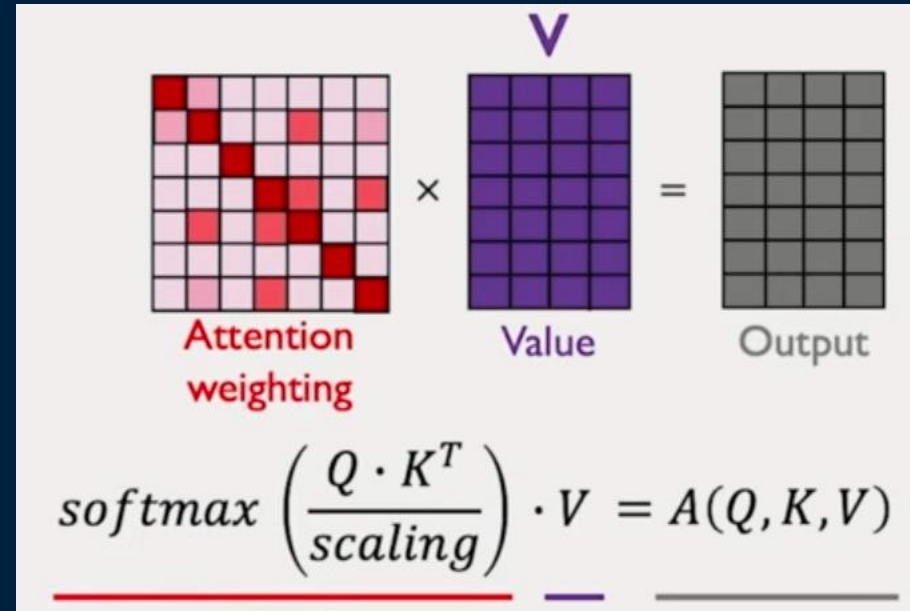
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Learning Self-Attention with NN

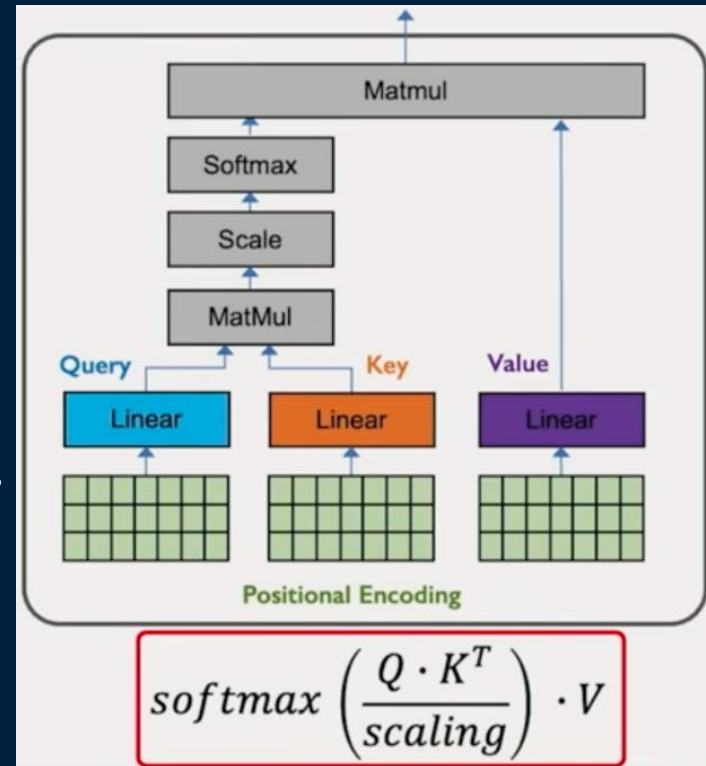
1. Encode position information
2. Extract query, key, value for each token
3. Compute attention weights
4. Extract features with high attention



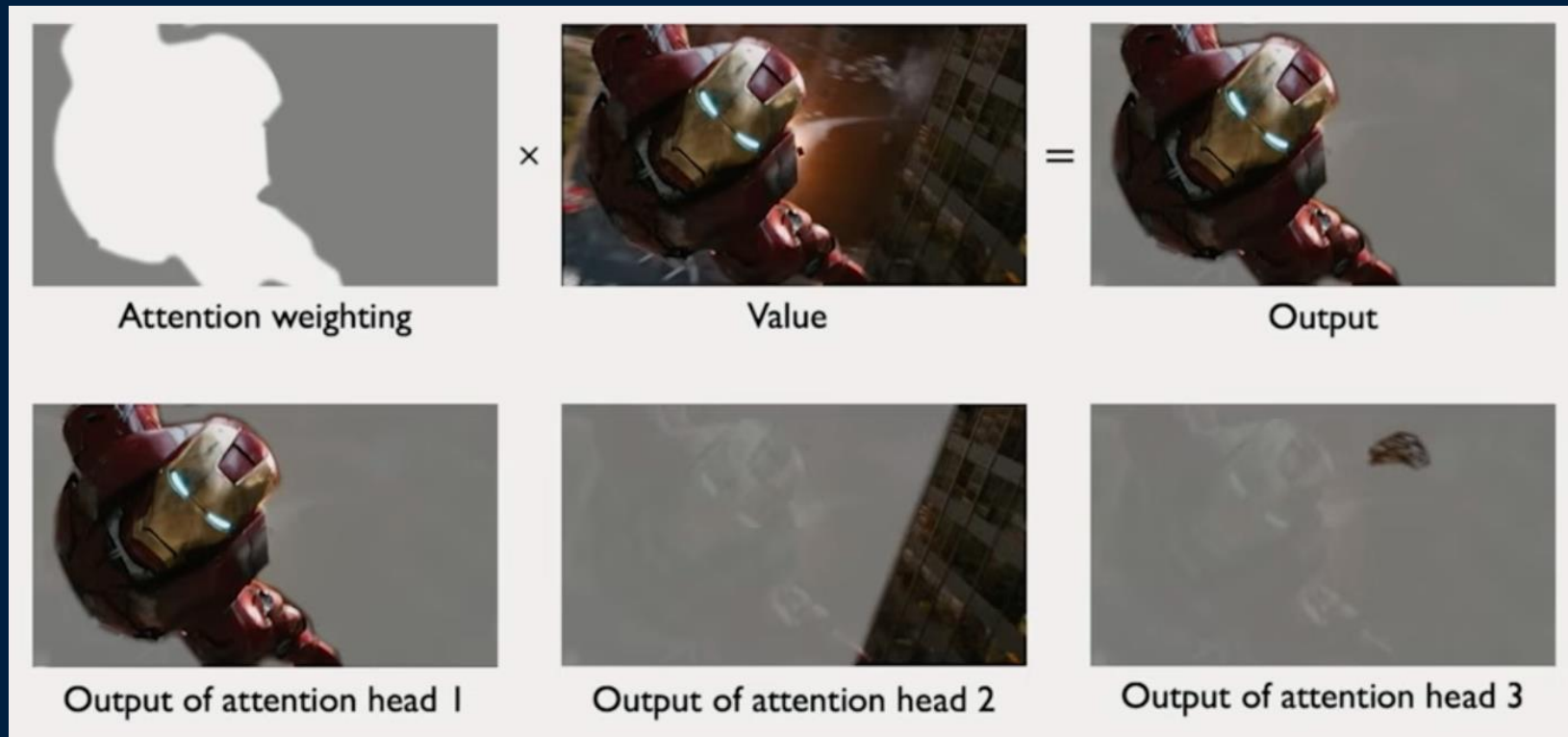
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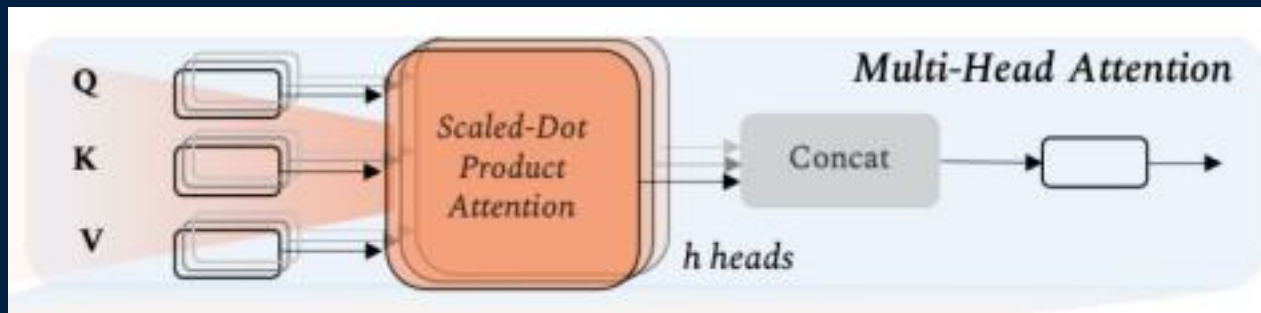
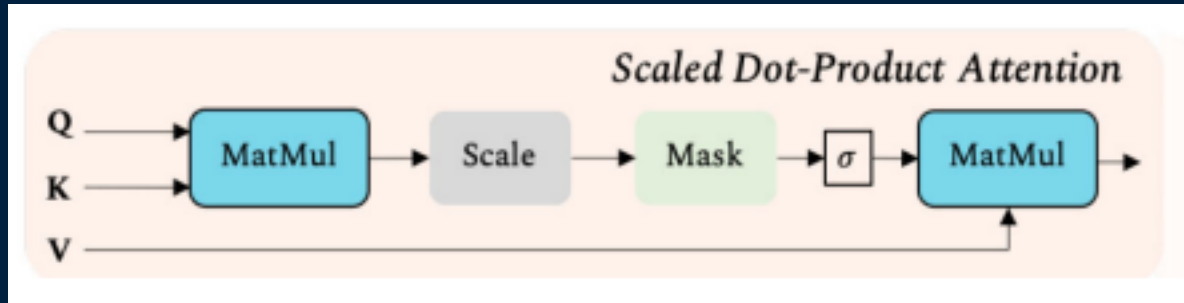
These operations form a **self-attention head** that can plug into a larger network. Each head attends to a different part of input.



Applying multiple self-attention heads



Multi-Head Attention



Question

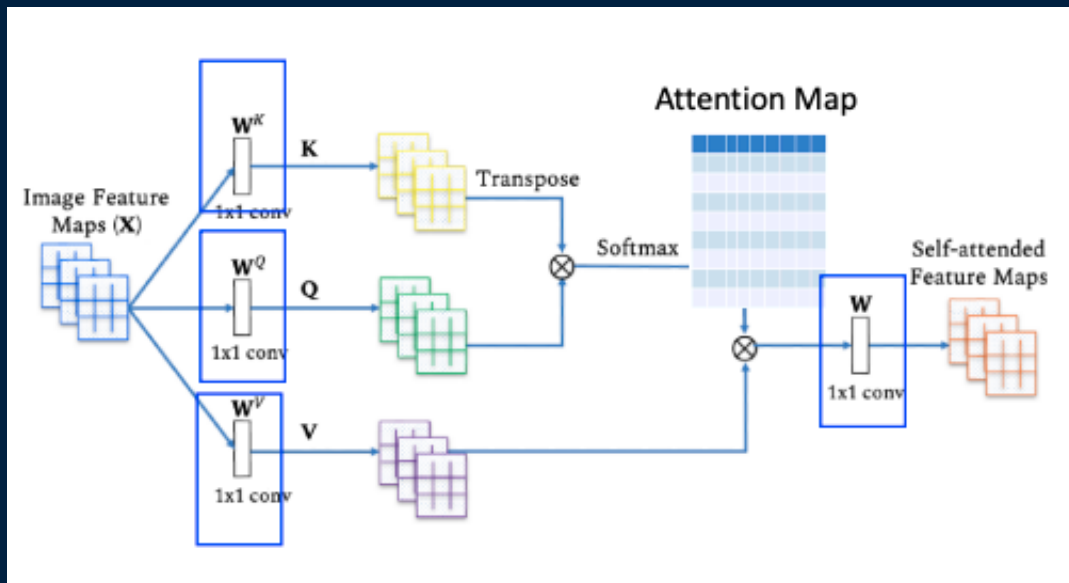
Question: Why do we need positional encoding in transformers?

- A. Because transformers process tokens one by one in sequence
- B. Because transformers need help understanding the order or position of tokens
- C. Because positional encoding reduces the number of parameters
- D. Because it replaces the value vectors
- E. Because it makes the model convolutional

Self Attention for Image

Self Attention for Image

- The **Queries** represent *what each location is looking for*.
- The **Keys** represent *what each location offers*.
- The **Values** represent *the information that will be shared*.



Question

Question: Which of the following statements is **FALSE** about multi-head self-attention?

- A. Different heads can learn different relationships between tokens
- B. One head may focus more on local structure while another may capture long-range interactions
- C. All heads are forced to learn the same attention pattern
- D. Multi-head attention lets the model attend in different learned subspaces
- E. Multi-head attention helps model several types of token relationships in parallel

Objectives

By the end of this lecture, students should be able to:

- Explain the main limitation of CNNs in modeling long-range dependencies
- Describe the intuition behind attention, query, key, and value
- Understand why positional information is needed in transformers
- Explain how self-attention computes relationships between tokens
- Describe the role of multi-head attention in learning different interaction patterns
- Explain the main components of the transformer encoder
- Understand how Vision Transformers convert image patches into tokens
- Describe the role of patch embeddings, positional embeddings, and the CLS token in ViT



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